Task 1 – Tagging Summary Report

Objective:
To structure and analyze free-text maintenance records by mapping complaints, causes, and corrections using the provided taxonomy. The goal was to identify recurring fault patterns and standardize data for clearer analysis.

Approach

- Reviewed each record's *Complaint*, *Cause*, and *Correction* to assign the most logical tags.
- Tagged data using the taxonomy categories:
  - **Root Cause:** Defined reason behind the issue (e.g., Poor Material, Faulty, Not Tightened).
  - **Symptom Condition:** Described what was observed (e.g., Leaking, Not Working, Loose).
  - **Fix Condition & Component:** Captured the corrective action taken (e.g., Replaced Sensor, Repaired Hose).
- Used **"Not Mentioned"** when the component was missing from taxonomy and added short comments for clarity.

Data Preparation

- Standardized inconsistent text (case, spacing, and formatting).
- Ensured all corrections matched their related fixes for tagging accuracy and consistency.
- Focused on creating structured data that could support trend analysis.

Key Observations

- **Frequent issues:** Leaks, loose fittings, and sensor faults.
- **Main causes:** Improper installation, poor-quality materials, or missing calibration.
- **Common fixes:** Replacement, repair, and calibration.

Insights

- Most recurring faults were **assembly-related and preventable** through better inspection and torque verification.
- Structured tagging helped reveal patterns that can drive **quality improvements** and reduce repetitive service incidents.
- Consistent documentation enhances traceability and helps identify failure trends faster.

Conclusion

The structured tagging converted unorganized service data into measurable categories, enabling quick identification of high-frequency issues and supporting data-driven decisions for process optimization.